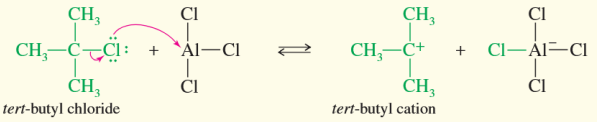
The Friedel–Crafts Alkylation

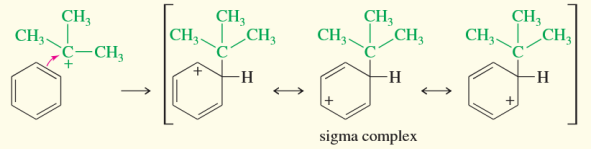
$R—X + AlCl_3 \rightleftharpoons R^+ + X—\bar{A}lCl_3$

(R is secondary or tertiary) — reacting electrophile

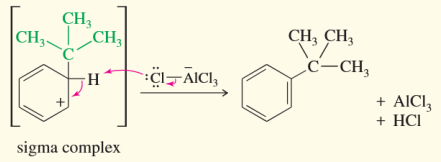
Step 1: Formation of a carbocation.

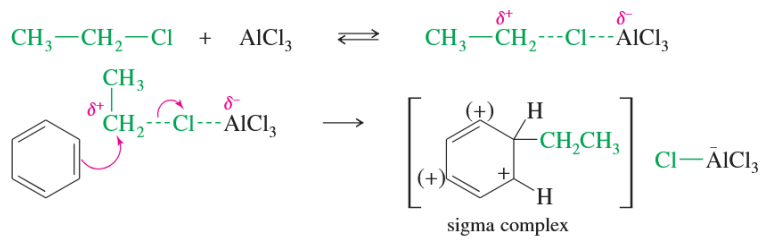


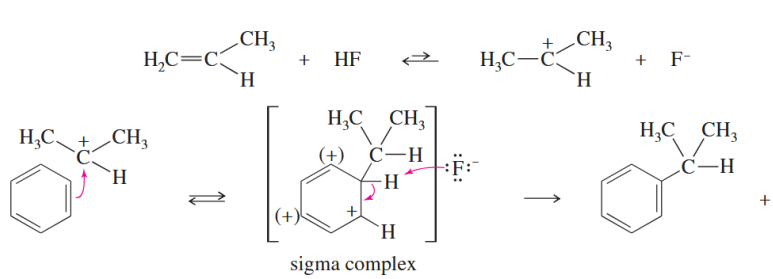
Step 2: Electrophilic attack forms a sigma complex.

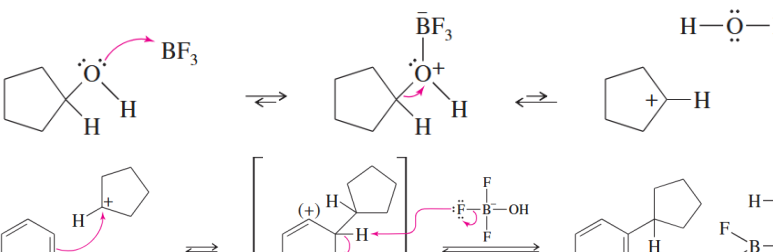


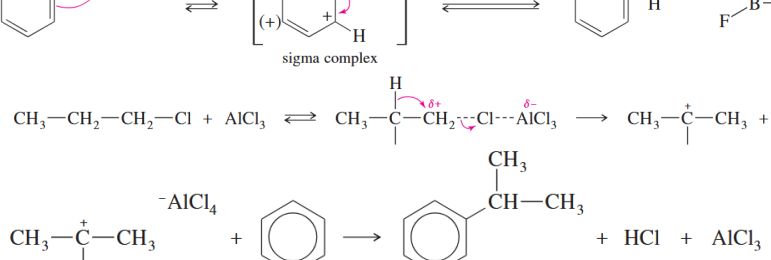
Step 3: Loss of a proton regenerates the aromatic ring and gives the alkylated product.

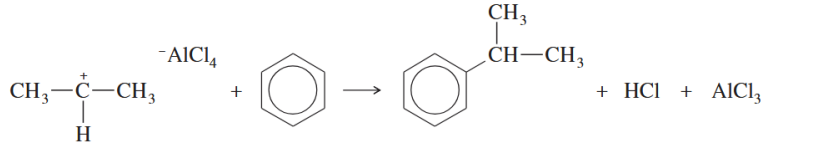


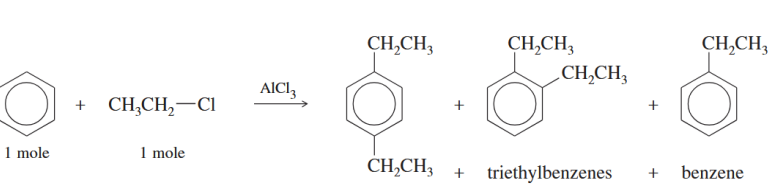




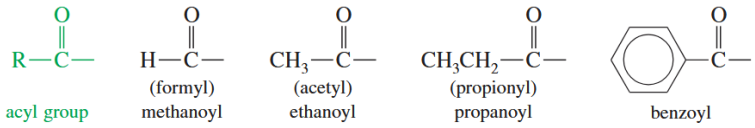




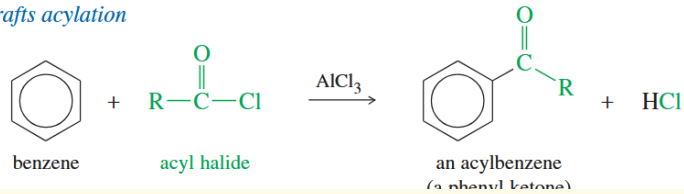




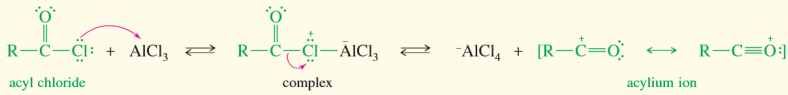
The Friedel–Crafts Acylation



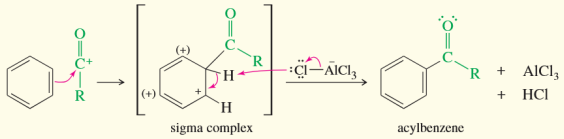
Friedel–Crafts acylation



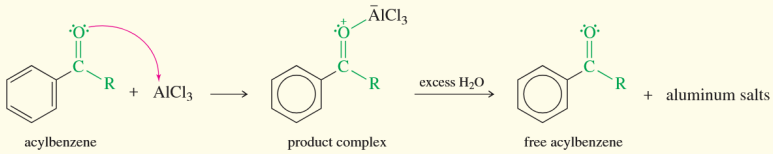
Step 1: Formation of an acylium ion.

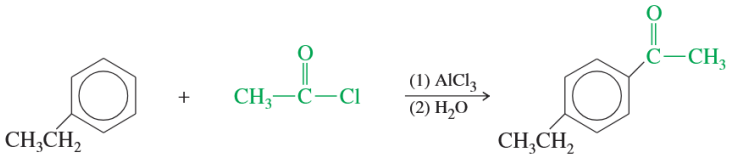


Steps 2 and 3: Electrophilic attack forms a sigma complex, and loss of a proton regenerates the aromatic system.



Step 4: Complexation of the product. The product complex must be hydrolyzed (by water) to release the free acylbenzene.

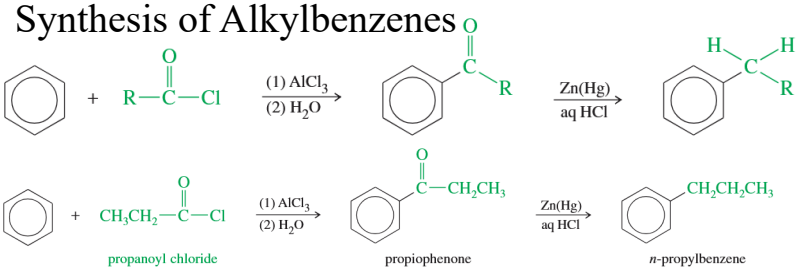


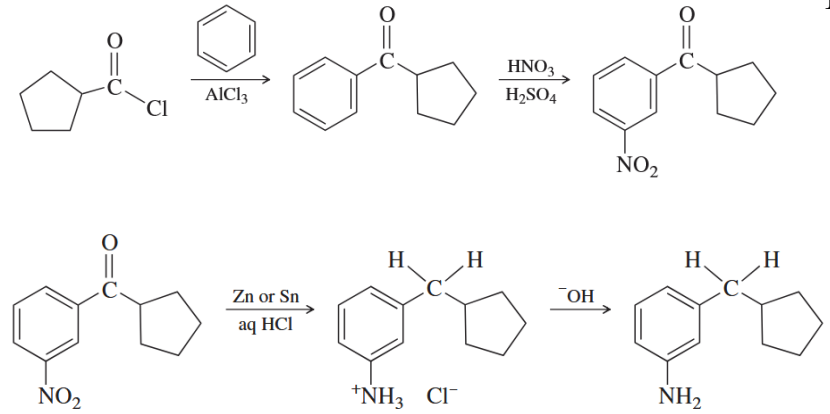


SUMMARY Comparison of Friedel–Crafts Alkylation and Acylation

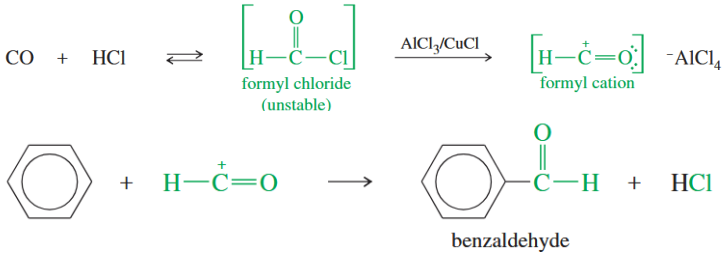
Alkylation	*Acylation*
The alkylation cannot be used with strongly deactivated derivatives.	Also true: Only benzene, halobenzenes, and activated derivatives are suitable.
The carbocations involved in the alkylation may rearrange.	Resonance-stabilized acylium ions are not prone to rearrangement.
Polyalkylation is commonly a problem.	The acylation forms a deactivated acylbenzene, which does not react further.

The Clemmensen Reduction: Synthesis of Alkylbenzenes

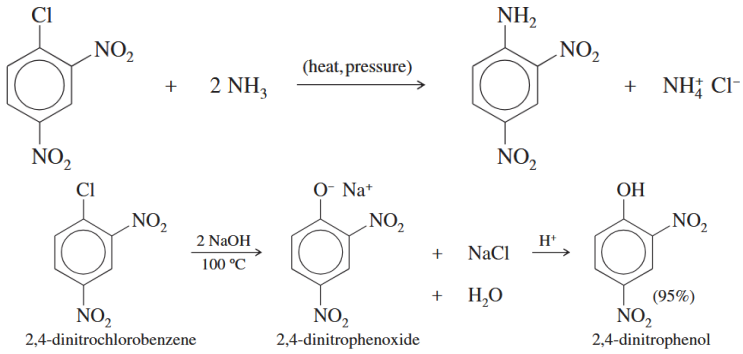




The Gatterman–Koch Formylation: Synthesis of Benzaldehyde

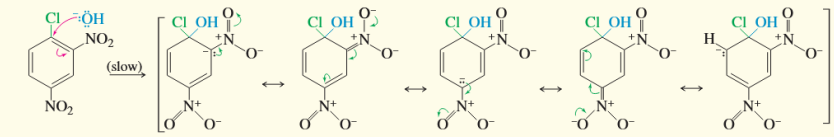


Nucleophilic Aromatic Substitution

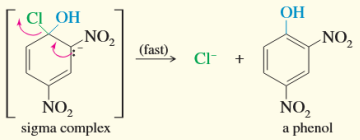


The Addition–Elimination Mechanism

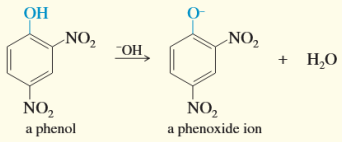
Step 1: Attack by the nucleophile gives a resonance-stabilized sigma complex.



Step 2: Loss of the leaving group gives the product.

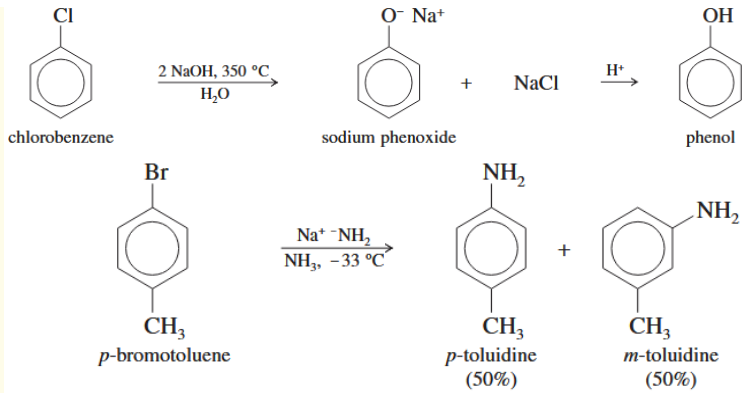


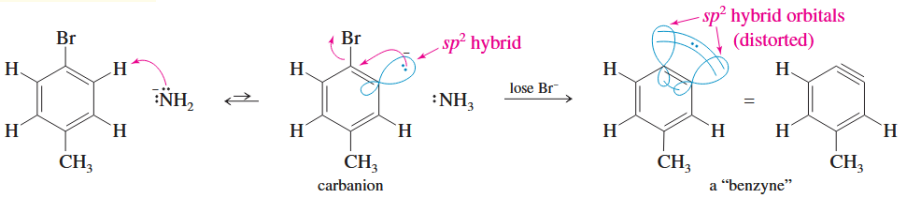
Step 3: This product (a phenol) is acidic, and is deprotonated by the base.

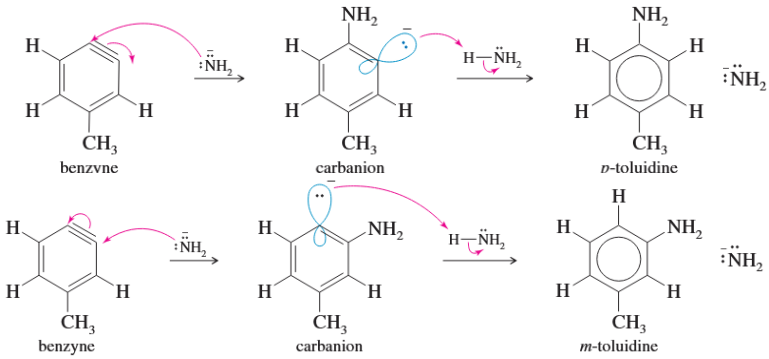


After the reaction is complete, acid would be added to reprotonate the phenoxide ion to give the phenol.

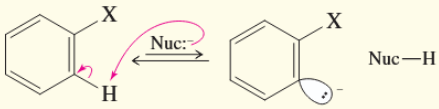
The Benzyne Mechanism: Elimination–Addition



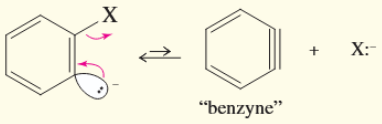




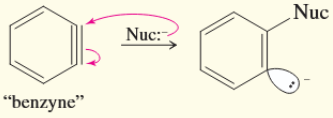
Step 1: Deprotonation adjacent to the leaving group gives a carbanion.



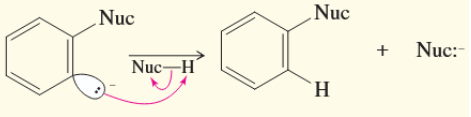
Step 2: The carbanion expels the leaving group to give a "benzyne" intermediate.



Step 3: The nucleophile attacks at either end of the reactive benzyne triple bond.



Step 4: Reprotonation gives the product.



Aromatic Substitutions Using Organometallic Reagents

Making the organolithium reagent:

$$R{-}X + 2\,Li \longrightarrow R{-}Li + LiX$$

(R = alkyl, vinyl, aryl)

Making the lithium organocuprate:

$$2\,R{-}Li + CuI \longrightarrow R_2CuLi + LiI$$

Coupling of the organocuprate with an alkyl, vinyl, or aryl halide:

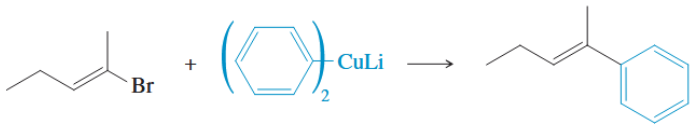
$$R'{-}X + R_2CuLi \longrightarrow R'{-}R + R{-}Cu + LiX$$

Examples:

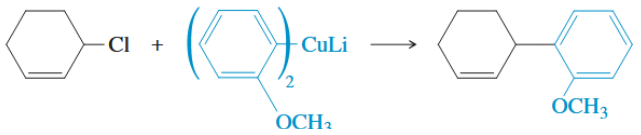
1. An aryl halide with an alkyl or vinyl cuprate.

iodobenzene — lithium divinylcuprate — styrene

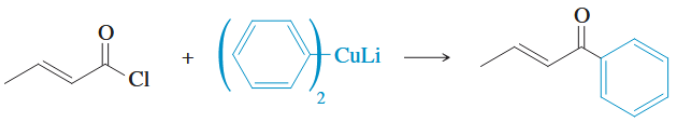
2. A vinyl halide with an aryl cuprate, preserving the stereochemistry of the vinyl halide.



3. An alkyl halide with an aryl cuprate.

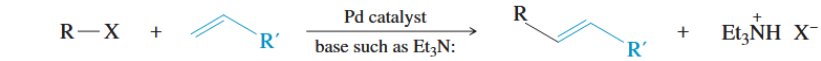


4. An acyl halide with an organocuprate, giving a ketone.



The Heck Reaction

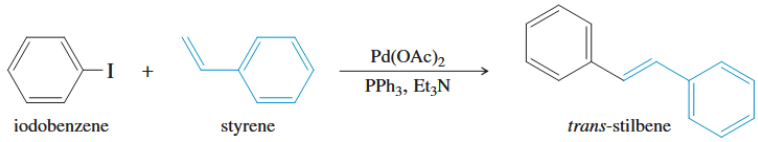
The Heck Reaction



(R = aryl or vinyl)
(X = Br or I)

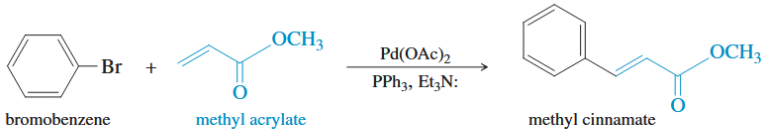
Examples of the Heck reaction:

1. An aryl halide with an aryl olefin.



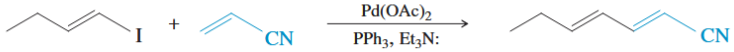
iodobenzene — styrene — *trans*-stilbene

2. An aryl halide with an conjugated acid or ester.



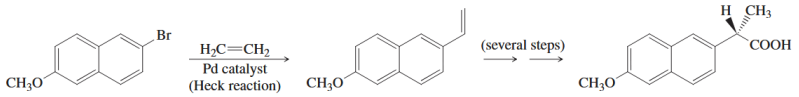
bromobenzene — methyl acrylate — methyl cinnamate

3. A vinyl halide with a conjugated nitrile.

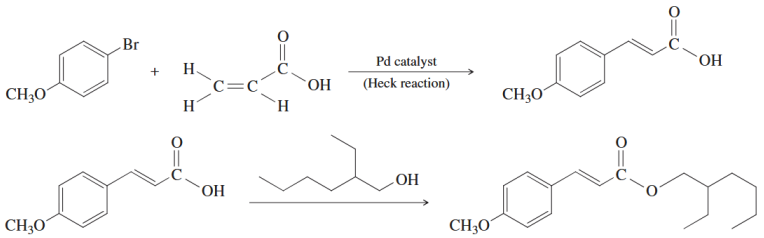


(*E*)-1-iodobut-I-ene — acrylonitrile — (2*E*, 4*E*)-hepta-2,4-dienenitrile

4. The industrial synthesis of naproxen, an over-the-counter analgesic and anti-inflammatory drug.



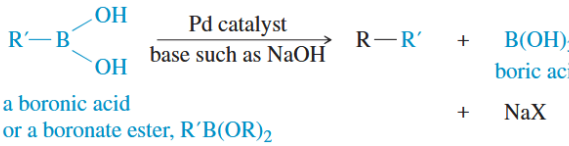
5. The industrial synthesis of octyl methoxycinnamate, a common ingredient in sunscreens.



"octyl methoxycinnamate"
(*E*)-2-ethylhexyl-4-methoxycinnamate

The Suzuki Reaction (Suzuki Coupling)

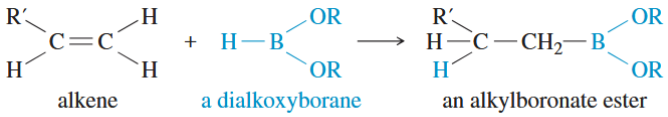
The Suzuki Reaction (Suzuki Coupling)



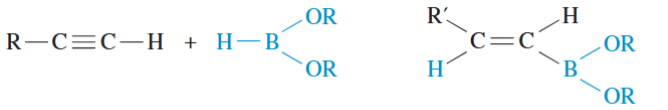
(R = aryl or vinyl)
(X = Br, I, or Cl)

a boronic acid
or a boronate ester, $R'B(OR)_2$

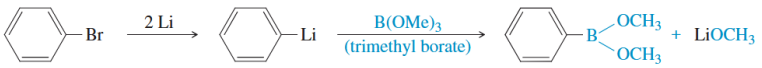
boric acid



alkene — a dialkoxyborane — an alkylboronate ester

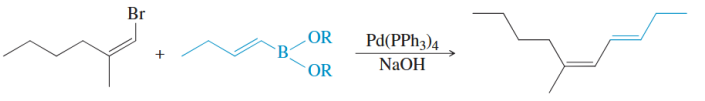


alkyne — a dialkoxyborane — a trans alkenylboronate ester



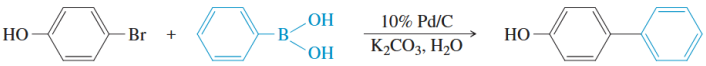
bromobenzene — phenyllithium — an arylboronate ester (dimethyl phenylboronate)

1. A vinyl halide with an alkenylboronate ester, preserving the stereochemistry of the reagents.



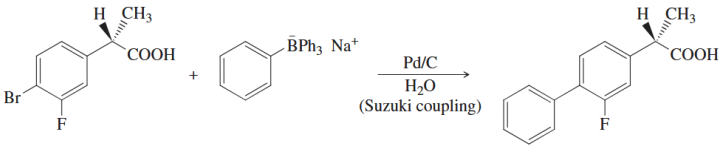
(*Z*)-vinyl halide — (*E*)-alkenylboronate ester — diene with (*E*, *Z*) stereochemistry

2. An aryl halide with an arylboronic acid, using palladium on carbon and water as the solvent.



p-bromophenol — phenylboronic acid — *p*-phenylphenol

3. Synthesis of the anti-inflammatory drug flurbiprofen by an environmentally efficient Suzuki coupling that uses water as the solvent and palladium on carbon as a reusable catalyst.

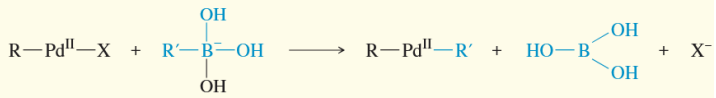


sodium tetraphenylborate — flurbiprofen

Step 1: Oxidative addition to the Pd gives Pd a higher oxidation number.

$$R{-}X + Pd^0 \longrightarrow R{-}Pd^{II}{-}X$$

Step 2: Exchange of ligands on Pd occurs.



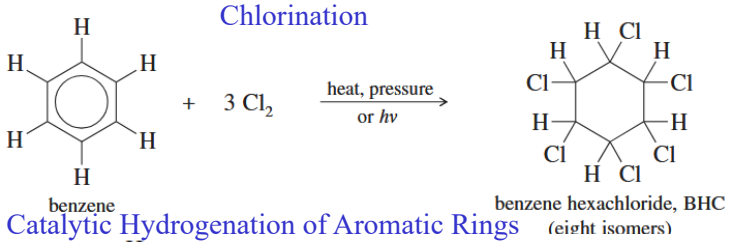
formed by addition of
⁻OH to $R'B(OH)_2$

Step 3: Reductive elimination from the Pd gives Pd a lower oxidation number.

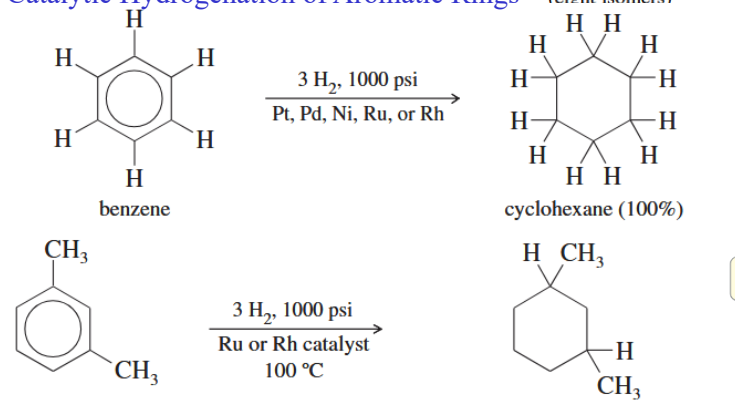
$$R{-}Pd^{II}{-}R' \longrightarrow R{-}R' + Pd^0$$

Addition Reactions of Benzene Derivatives

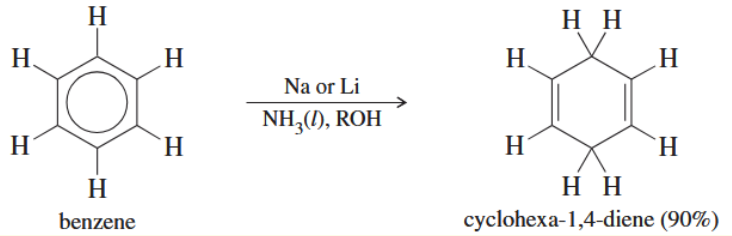
Chlorination



Catalytic Hydrogenation of Aromatic Rings



Birch Reduction

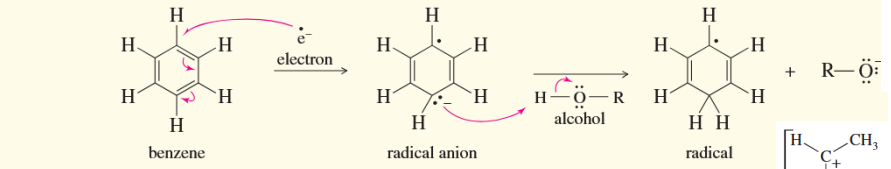


Preceding step: Formation of solvated electrons in the ammonia solution.

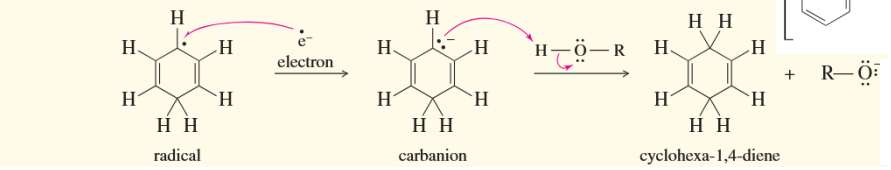
$$NH_3 + Na \rightleftharpoons NH_3 \cdot e^- \text{ (deep blue solution)} + Na^+$$

solvated electron

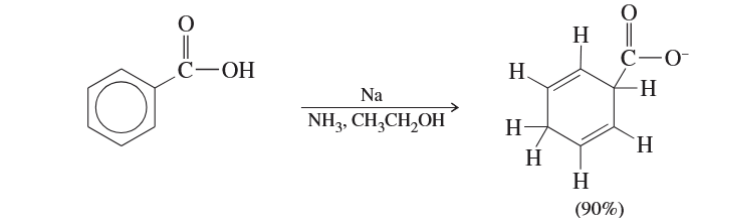
Steps 1 and 2: Addition of an electron, followed by a proton, forms a radical.



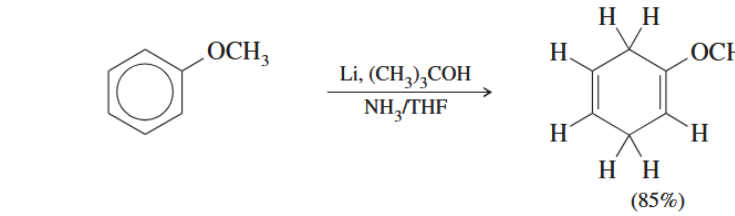
Steps 3 and 4: Addition of a second electron, followed by a proton, gives the product.



A carbon bearing an electron-withdrawing carbonyl group is reduced

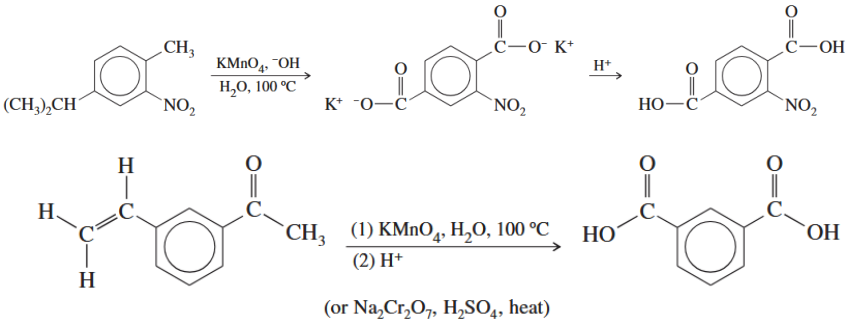


A carbon bearing an electron-releasing alkoxy group is not reduced

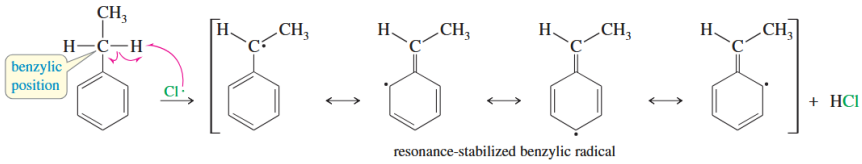


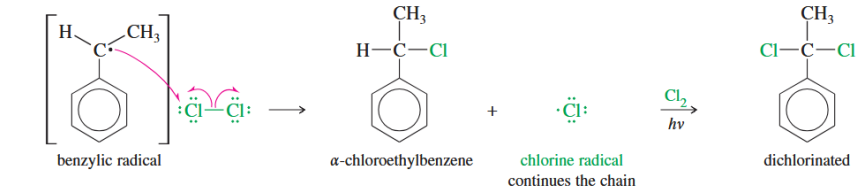
Side-Chain Reactions of Benzene Derivatives

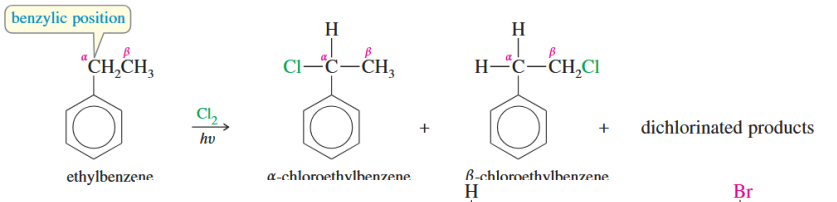
Permanganate Oxidation

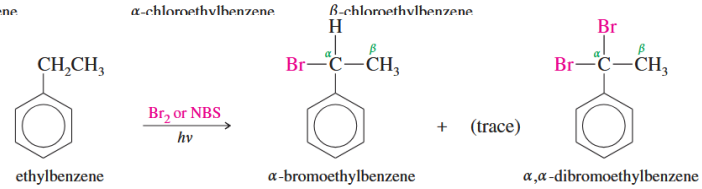


Side-Chain Halogenation

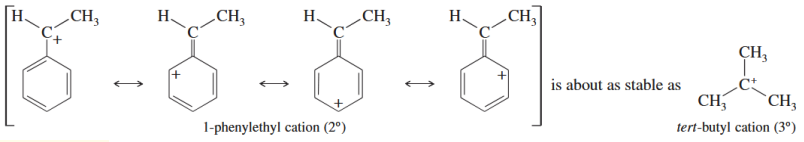


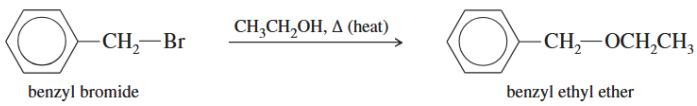


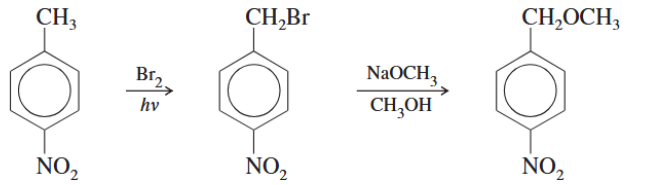


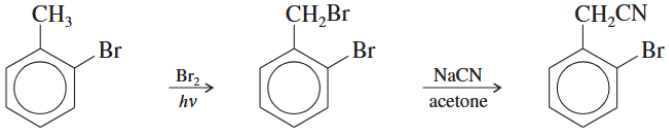


Nucleophilic Substitution at the Benzylic Position

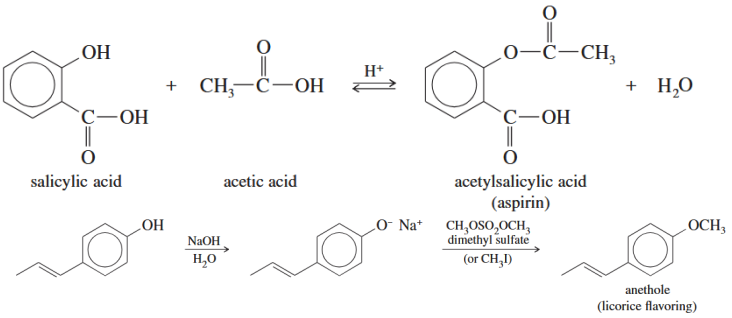




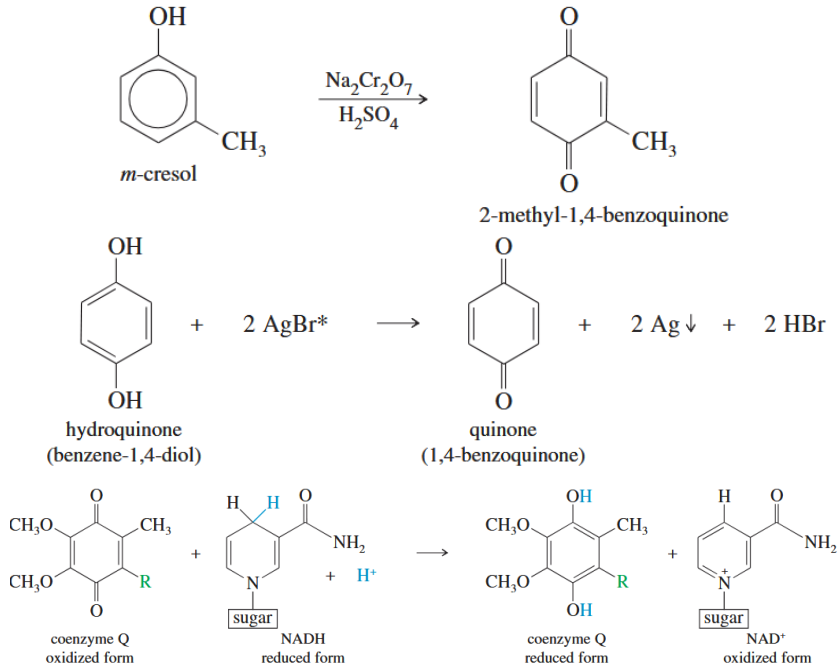




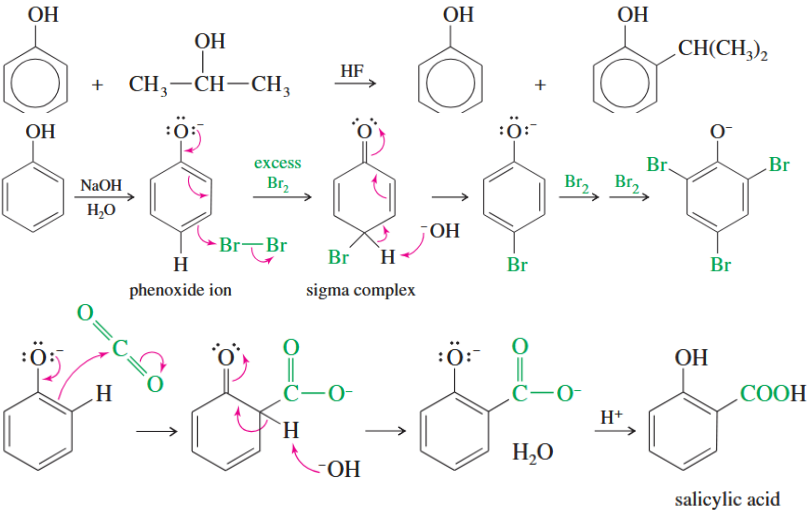
Reactions of Phenols



Oxidation of Phenols to Quinones



Electrophilic Aromatic Substitution of Phenols

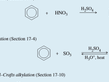


Reactions of Aromatic Compounds

1. Electrophilic aromatic substitution

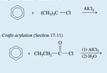
a. *Halogenation* (Section 17-2)

b. *Nitration* (Section 17-3)

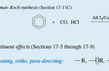


Nitration followed by reduction gives anilines.

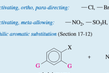
c. *Sulfonation* (Section 17-4)



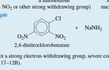
d. *Friedel–Crafts alkylation* (Section 17-10)



e. *Friedel–Crafts acylation* (Section 17-11)



f. *Gattermann–Koch methods* (Section 17-11C)



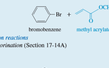
g. *Substituent effects* (Section 17-5 through 17-9)

Activating, ortho, para-directing: —R, —OR, —NH2, —NR2, —C6H5 (amines, anisides)

Deactivating, ortho, para-directing: —Cl, —Br, —I

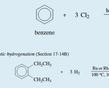
Deactivating, meta-allowing: —NO2, —SO3H, —NR3+, —C≡O, —C≡N

2. Nucleophilic aromatic substitution (Section 17-12)



(G = NO_2 or other strong withdrawing group)

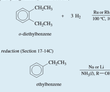
Example:



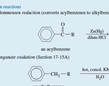
If G is not a strong electron-withdrawing group, severe conditions are required, and a benzyne mechanism is involved (Section 17–12B).

3. Organometallic Couplings

a. *Organocuprate couplings* (Section 17-13A)

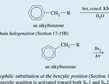


b. *Heck coupling* (Section 17-13B)

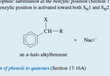


4. Addition reactions

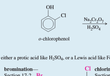
a. *Chlorination* (Section 17-14A)



b. *Catalytic hydrogenation* (Section 17-14B)

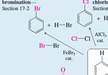


c. *Birch reduction* (Section 17-14C)

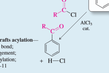


5. Side-chain reactions

a. *The Clemmensen reduction* (converts acylbenzenes to alkylbenzenes, Section 17-11B)



b. *Permanganate oxidation* (Section 17-15A)

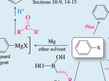


Halogen, $-NO_2$, and $-SO_3H$ groups survive this benzal oxidation.

c. *Side-chain halogenation* (Section 17-15B)

d. *Nucleophilic substitution at the benzylic position* (Section 17-15C)

The benzylic position is activated toward both S_N1 and S_N2 displacements.



6. Oxidation of phenols to quinones (Section 17-16A)

A catalyst, either a protic acid like H_2SO_4, or a Lewis acid like $FeBr_3$ or $AlCl_3$, is needed for each reaction.

GUIDE TO ORGANIC REACTIONS IN CHAPTER 17

Reactions covered in Chapter 17 are shown in red. Reactions covered in earlier chapters are shown in blue.

Substitution	Addition	Elimination	Oxidation/Reduction
Nucleophilic • S_N1 (Sec. 6.12) • S_N2 (Sec. 6.12) • sp^2 C (Ben. Arom. Subst.) Ch 17, 19 • sp^2 C (Carbonyl Acyl Subst.) Ch 11, 20, 21, 22	**Nucleophilic** • sp^3 C (Sec. 10.8, 21.7) • sp^2 C (Conjugate adds) Ch 22	**Basic conditions (E2)** • E2 dehydrohalogenation Ch 5 • E2 alcohol dehydration Ch 11 • E1 dehydrohalogenation Ch 10 • E1 dehydration of alcohols Ch 11 **Acidic conditions (E1)** • E1 dehydrohalogenation Ch 5 • E1 dehydration of alcohols Ch 11 • Pericyclic (Cope elimination) Ch 19	**Oxidation** • Epoxidation Ch 11 • Ozonolysis oxidative cleavage Ch 5, 11, 17, 22 • Oxidation of benzylic groups Ch 11, 18, 19, 20 **Reduction** • Hydride reduction Ch 5, 10, 11, 14, 18, 20, 21 • Catalytic hydrogenation Ch 5, 11, 16, 19 **metal** Ch 5, 17, 18, 19
Electrophilic • sp^2 C (Alkene halogenation) Ch 5, 11, 17 • sp^2 C (Sulfonyl rxn) Ch 19	**Electrophilic** • sp^2 C (Elect. Adds.) Ch 5, 6, 11, 17 • sp^2 C (Carbene Adds.) Ch 5		
Radical • sp^3 C (Halogenation) Ch 5, 11, 17	**Radical** • sp^2 C (HBr + ROOR) Ch 5		
Organometallic Ch 5, 17 **Suzuki** Ch 17 **Heck** Ch 17	**Pericyclic** • cycloaddition (Diels Alder) Ch 22 **Oxidation** • epoxidation Ch 11, 22 **Reduction** • hydrogenation Ch 5, 11, 16, 19		

Essential Terms

activating group — A substituent that makes the aromatic ring more reactive (usually toward electrophilic aromatic substitution) than benzene. (p. 817)

acyl cation; acylium ion — An sp^2 carbocation in which an acyl group attached. (p. 855)

allyl cation — A carbocation in which a carbon bonded to a vinyl group, $R_2C{=}CR{-}C^+$ An sp^2 group fragment with a positive charge. (p. 855)

allylic — Adjacent to an sp^3 substituent consisting of an sp^2 group bonded through an oxygen atom, $-O-$ or $-S-$. (p. 818)

benzylic position — The carbon atom of an sp^3 group that is directly bonded to a benzene ring; the position a to a benzene ring. (p. 853)

benzyne — A reactive intermediate in some nucleophilic aromatic substitutions, benzyne is benzene with two hydrogen atoms removed. It can be drawn with a highly strained triple bond in the six-membered ring. (p. 842)

Birch reduction — The partial reduction of a benzene ring by sodium or lithium in liquid ammonia. The products are usually cyclohexa-1,4-dienes. (p. 850)

Clemmensen reduction — The reduction of a carbonyl group to a methylene group by zinc amalgam, Zn(Hg), in dilute hydrochloric acid. (p. 837)

amalgam; deactivating group — An alloy of a metal with mercury.

electrophilic aromatic substitution — A substituent that makes the aromatic ring less reactive (usually toward electrophilic aromatic substitution) than benzene. (p. 823)
(**EAS**) Replacement of a hydrogen on an aromatic ring by a strong electrophile. (p. 810)

Friedel–Crafts acylation — Formation of an arylketone by substitution of an acylium ion on an aromatic ring. (p. 835)

Friedel–Crafts alkylation — Formation of an alkyl-substituted benzene derivative by substitution of an alkyl carbocation or carbocation-like species on an aromatic ring. (p. 830)

Gattermann–Koch synthesis — The synthesis of benzaldehydes by treating a benzene derivative with CO and HCl using an $AlCl_3/CuCl$ catalyst. (p. 838)

halonium ion — A positively charged ion that has a positive charge (or partial positive charge) on a halogen atom. Typically, in a halonium ion the halogen atom has two bonds and bears a formal plus charge. (Specific: chloronium ion, bromonium ion, and so forth.) (p. 826)

Heck reaction — A palladium-catalyzed coupling of an aryl or vinyl halide with an alkene to form a new C—C bond at the less substituted end of the alkene. (p. 895)

inductive stabilization — Stabilization of a reactive intermediate by donation or withdrawal of electron density through sigma bonds. (p. 820)

meta director — (**meta-allowed**) A substituent that deactivates primarily the ortho and para positions, leaving the meta position the least deactivated and most reactive. (p. 823)

nitration — Replacement of a hydrogen atom by a nitro group, $-NO_2$. (p. 813)

nitronium ion — The NO_2^+ ion, $O{=}N{\equiv}O^+$. (p. 813)

nucleophilic aromatic substitution — (**NAS**) Replacement of a leaving group on an aromatic ring by a strong nucleophile. Usually takes place by an addition–elimination mechanism or by a benzyne mechanism. (p. 839)

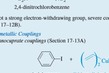
organocuprate — A compound containing alkyl–copper bonds, commonly of formula R_2CuLi (Gilman reagent). The nitrogen analogs (lithiodialkylcuprates or dialkylcuprates) displace an organohalogen bond to form a new C—C bond. (p. 895)

ortho, para director — A substituent that activates primarily the ortho and para positions toward attack. (p. 817)

quinone — A derivative of a cyclohexadiene-dione. Common quinones are the 1,4-quinones (para-quinones); the less stable 1,2-quinones (ortho-quinones) are relatively uncommon. (p. 857)

resonance stabilization — Stabilization of a reactive intermediate by donation or withdrawal of electron density through pi bonds. (p. 820)

resonance-donating; resonance-withdrawing — (**p-donating**) Capable of donating electrons through resonance involving pi bonds. (pp. 820, 826)
(**p-withdrawing**) Capable of withdrawing electron density through resonance involving pi bonds. (p. 822)



sigma complex — An intermediate in electrophilic aromatic substitution or nucleophilic aromatic substitution with a sigma bond between the electrophile or nucleophile and the former aromatic ring. The sigma complex bears a delocalized positive charge in electrophilic aromatic substitution and a delocalized negative charge in nucleophilic aromatic substitution. (p. 809)

sulfonation — Replacement of a hydrogen atom by a sulfonic acid group, $-SO_3H$. (p. 814)

desulfonation — Replacement of the $-SO_3H$ group by a hydrogen. With benzene derivatives, this is done by heating with water or steam and acid. (p. 816)

Suzuki reaction — (**Suzuki coupling**) A palladium-catalyzed substitution that couples an aryl or vinyl halide with a alkyl, aryl, or vinyl boronic acid or boronate ester. (p. 845)